

Logos Final Corrected Master Leaderboard

Summary

This corrected leaderboard separates Logos results into four rankings instead of forcing every model into one score. This is stricter and more defensible: pure quality, deployment efficiency, compact-model efficiency, and scaling trend answer different research questions.

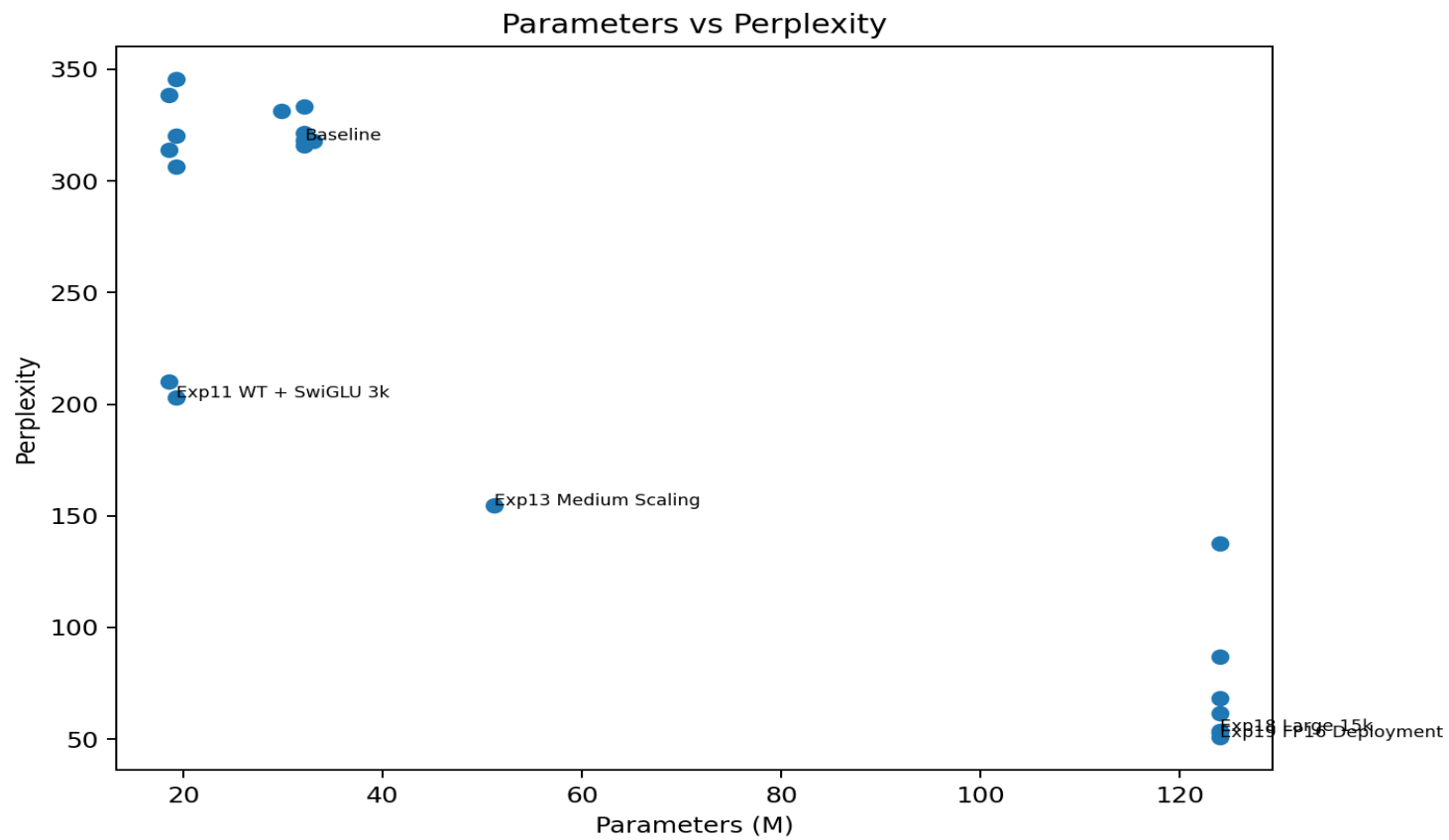
Headline conclusions

Question	Answer	Evidence
Best trained model	Exp18 Large 15k	124.03M params, PPL 53.57, val loss 3.9811.
Best deployment model	Exp19 FP16 Deployment	310.24 MB checkpoint, 5.96 GB VRAM, 9.14 ms/token, sampled PPL 50.89.
Best compact model	Exp12 Wider + WT + SwiGLU 3k	18.54M params, 70.76 MB checkpoint, PPL 210.00.
Strongest scaling evidence	Exp13 to Exp18	PPL improved from 154.79 to 53.57 as scale/training increased.

Ranking 1: Pure Quality - Trained Models Only

Compression rows are excluded here because their validation losses are sampled estimates. The correct pure-trained winner is Exp18.

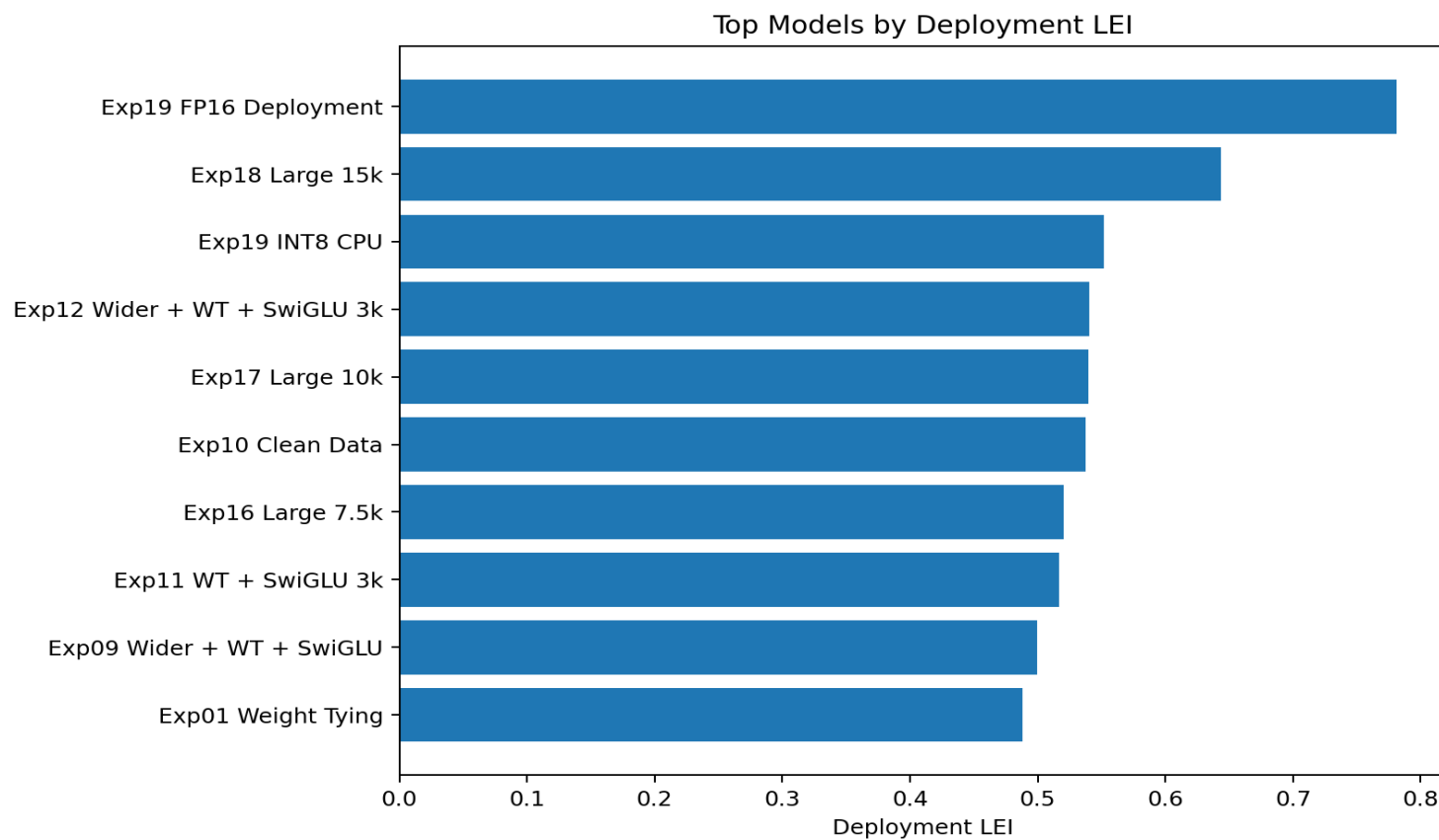
Experiment	Params M	Samples	Ctx	Iters	Val Loss	PPL	Ckpt MB	VRAM GB
Exp18 Large 15k	124.03	100000	512	15000	3.98	53.57	473.20	17.73
Exp17 Large 10k	124.03	100000	512	10000	4.12	61.36	473.20	17.73
Exp16 Large 7.5k	124.03	100000	512	7500	4.23	68.40	473.20	17.73
Exp15 Large 5k	124.03	100000	512	5000	4.47	87.03	473.20	17.73
Exp14 Large Scaling	124.03	100000	512	3000	4.92	137.60	473.20	17.73
Exp13 Medium Scaling	51.20	50000	512	3000	5.04	154.79	195.36	13.09
Exp11 WT + SwiGLU 3k	19.24	10000	256	3000	5.31	202.82	73.45	11.18
Exp12 Wider + WT + SwiGLU 3k	18.54	10000	256	3000	5.35	210.00	70.76	10.54
Exp05 WT + SwiGLU	19.24	10000	256	1500	5.72	306.36	73.44	11.18
Exp09 Wider + WT + SwiGLU	18.54	10000	256	1500	5.75	313.87	70.76	10.54
Exp04 SwiGLU	32.16	10000	256	1500	5.76	315.93	122.71	11.33
Exp08 Wider-Shallower	33.07	10000	256	1500	5.76	318.00	126.15	10.43



Ranking 2: Deployment Efficiency

This ranking includes compressed variants and focuses on approximate quality, checkpoint size, VRAM, generation latency, and parameter count. Exp19 FP16 is the strongest deployment result.

Experiment	Category	PPL	Ckpt MB	VRAM GB	ms/tok	Deploy LEI
Exp19 FP16 Deployment	compression	50.89	310.24	5.96	9.14	0.7808
Exp18 Large 15k	scaling	53.57	473.20	17.73	9.11	0.6434
Exp19 INT8 CPU	compression	52.77	267.05	N/A	20.34	0.5518
Exp12 Wider + WT + SwiGLU 3k	longer_train	210.00	70.76	10.54	12.06	0.5406
Exp17 Large 10k	scaling	61.36	473.20	17.73	14.77	0.5400
Exp10 Clean Data	data_test	338.53	70.76	10.54	8.65	0.5374
Exp16 Large 7.5k	scaling	68.40	473.20	17.73	11.92	0.5205
Exp11 WT + SwiGLU 3k	longer_train	202.82	73.45	11.18	14.03	0.5169
Exp09 Wider + WT + SwiGLU	small_efficiency	313.87	70.76	10.54	12.71	0.4996
Exp01 Weight Tying	small_ablation	320.32	73.44	10.69	12.64	0.4883
Exp05 WT + SwiGLU	small_combined	306.36	73.44	11.18	15.04	0.4732
Exp06 WT + SwiGLU ctx512	context_test	345.43	73.69	11.18	13.62	0.4720



Ranking 3: Compact Model Ranking

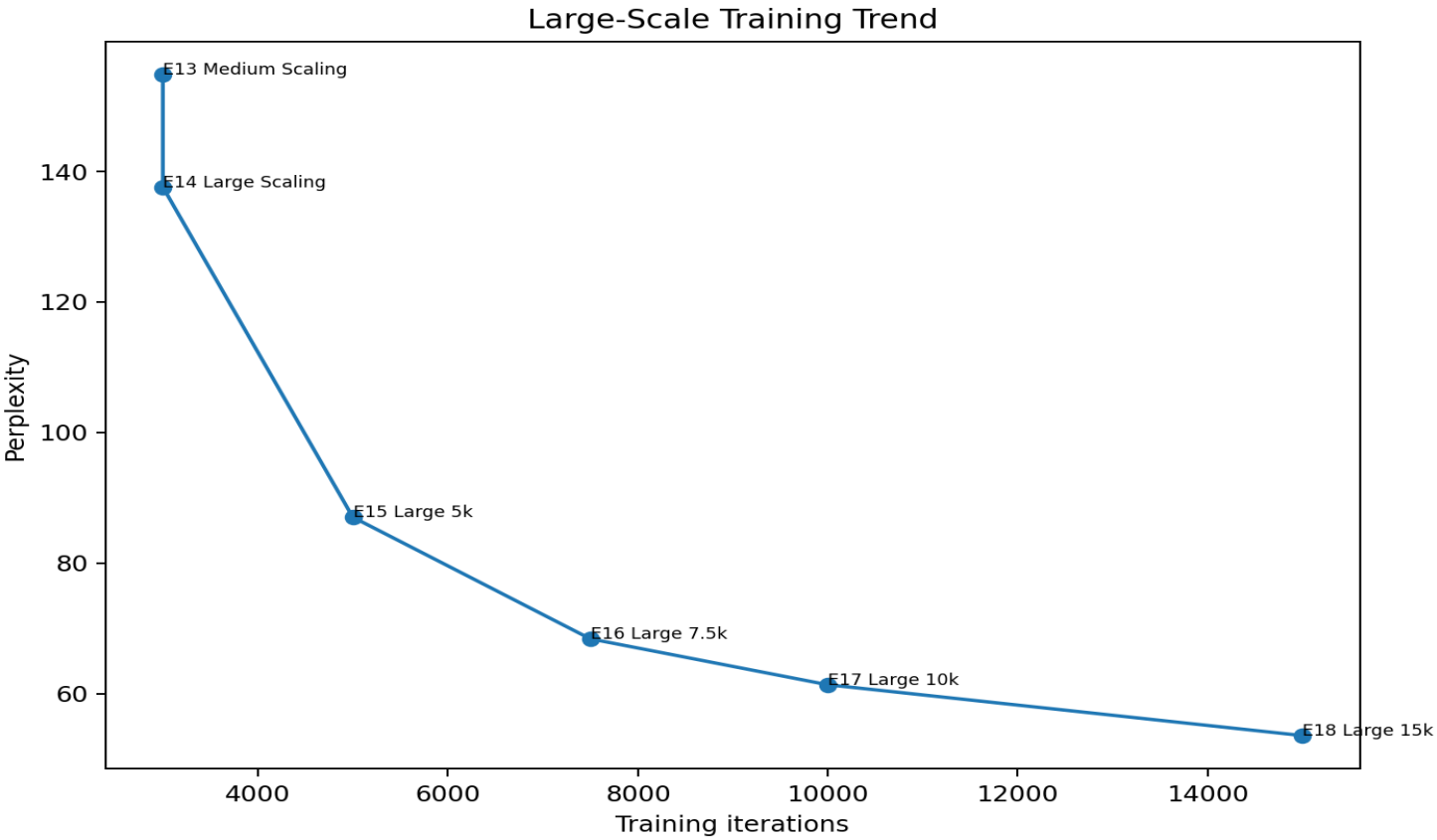
This section focuses on models below 25M parameters. Exp12 is the best compact balance after longer training, while Exp11 is slightly better in quality but larger/slower.

Experiment	Params M	Iters	Val Loss	PPL	Ckpt MB	ms/tok	Wh
Exp11 WT + SwiGLU 3k	19.24	3000	5.31	202.82	73.45	14.03	19.48
Exp12 Wider + WT + SwiGLU 3k	18.54	3000	5.35	210.00	70.76	12.06	16.29
Exp05 WT + SwiGLU	19.24	1500	5.72	306.36	73.44	15.04	10.49
Exp09 Wider + WT + SwiGLU	18.54	1500	5.75	313.87	70.76	12.71	9.06
Exp01 Weight Tying	19.24	1500	5.77	320.32	73.44	12.64	9.87
Exp10 Clean Data	18.54	1500	5.82	338.53	70.76	8.65	8.98
Exp06 WT + SwiGLU ctx512	19.31	1500	5.84	345.43	73.69	13.62	10.72

Ranking 4: Scaling Trend

The scaling phase is the strongest evidence that the Logos recipe improves with model scale, data scale, and training duration. Exp18 remains the best trained scaling result so far.

Experiment	Params M	Samples	Ctx	Iters	Val Loss	PPL	Ckpt MB	Wh
Exp11 WT + SwiGLU 3k	19.24	10000	256	3000	5.31	202.82	73.45	19.48
Exp13 Medium Scaling	51.20	50000	512	3000	5.04	154.79	195.36	29.70
Exp14 Large Scaling	124.03	100000	512	3000	4.92	137.60	473.20	59.76
Exp15 Large 5k	124.03	100000	512	5000	4.47	87.03	473.20	96.03
Exp16 Large 7.5k	124.03	100000	512	7500	4.23	68.40	473.20	141.11
Exp17 Large 10k	124.03	100000	512	10000	4.12	61.36	473.20	183.67
Exp18 Large 15k	124.03	100000	512	15000	3.98	53.57	473.20	277.39



Methodological Notes and Limitations

1. LEI rankings are weight-dependent. They should be presented as a resource-aware decision tool, not as a universal scientific law.
2. Experiment 19 FP16 and INT8 results use sampled validation estimates; therefore, FP16 should not be claimed as a better trained model than Exp18. It is a deployment-compressed version that preserved approximate quality.
3. The strongest novelty remains the controlled resource-aware framework and scaling validation, not a new Transformer block or a patent-level algorithm.
4. The next technical step should be a formal resource-aware selection algorithm or an adaptive training policy, because that would move the project closer to genuine method novelty.

Appendix: Full Ranked Table

Experiment	Category	Params M	PPL	Ckpt MB	VRAM GB	ms/tok	Wh	Deploy LEI
Exp19 FP16 Deployment	compression	124.03	50.89	310.24	5.96	9.14	N/A	0.7808
Exp19 INT8 CPU	compression	124.03	52.77	267.05	N/A	20.34	N/A	0.5518
Exp18 Large 15k	scaling	124.03	53.57	473.20	17.73	9.11	277.39	0.6434
Exp17 Large 10k	scaling	124.03	61.36	473.20	17.73	14.77	183.67	0.5400
Exp16 Large 7.5k	scaling	124.03	68.40	473.20	17.73	11.92	141.11	0.5205
Exp15 Large 5k	scaling	124.03	87.03	473.20	17.73	16.76	96.03	0.4206
Exp14 Large Scaling	scaling	124.03	137.60	473.20	17.73	15.13	59.76	0.3313
Exp13 Medium Scaling	scaling	51.20	154.79	195.36	13.09	13.19	29.70	0.3952
Exp11 WT + SwiGLU 3k	longer_train	19.24	202.82	73.45	11.18	14.03	19.48	0.5169
Exp12 Wider + WT + SwiGLU 3k	longer_train	18.54	210.00	70.76	10.54	12.06	16.29	0.5406
Exp05 WT + SwiGLU	small_combined	19.24	306.36	73.44	11.18	15.04	10.49	0.4732
Exp09 Wider + WT + SwiGLU	small_efficiency	18.54	313.87	70.76	10.54	12.71	9.06	0.4996
Exp04 SwiGLU	small_ablation	32.16	315.93	122.71	11.33	7.78	11.37	0.4455
Exp08 Wider-Shallower	shape_test	33.07	318.00	126.15	10.43	8.53	9.63	0.4347
Baseline	baseline	32.16	318.42	122.71	10.84	10.20	10.78	0.4129
Exp01 Weight Tying	small_ablation	19.24	320.32	73.44	10.69	12.64	9.87	0.4883
Exp03 RMSNorm	small_ablation	32.16	321.24	122.69	11.11	13.51	11.50	0.3823
Exp07 Deeper-Narrower	shape_test	29.88	331.29	114.02	11.11	13.65	11.97	0.3902
Exp02 GELU	small_ablation	32.16	333.33	122.71	11.10	14.45	10.84	0.3741
Exp10 Clean Data	data_test	18.54	338.53	70.76	10.54	8.65	8.98	0.5374
Exp06 WT + SwiGLU ctx512	context_test	19.31	345.43	73.69	11.18	13.62	10.72	0.4720